

# Beaglebone Black: Flashing eMMC Through an SD Card

The Beaglebone Black is a low-power open source single-board computer. eMMC is short for 'embedded multi media controller' which refers to a package integrating Flash memory and a memory controller in the same silicon die. In this article, we flash the eMMC on to the Beaglebone Black using an SD card.



The Beaglebone Black Rev C features a Sitara AM3358BZ100, 1GHz processor and 512MB DDR3L RAM. Its significant features are lower costs, and an increase in both performance and memory size. In this low power device, EEPROM has been reduced from 32KB to 4KB, so that the cost falls. The board features a 4GB NAND (eMMC), and has an HDMI and a USB interface. In addition, it has a GPIO3\_21 with a 24.576MHz clock. It also carries a microSD card. On-board Flash memory of 2GB acts as the hard drive for the board to host a Linux operating system and other software development tools.

With a user-friendly, browser-based Bonescript programming environment called Cloud9, a learner can easily program the Beaglebone Black (BBB) board to rapidly prototype electronic systems that interface with real-world applications. As user knowledge develops, the board provides more complicated interfaces including C/C++ functions to access digital and analogue pins aboard the ARM Cortex A8 microprocessor. The full power and capability of the BBB board can be programmed in the underlying onboard Linux operating system, such as Angstrom or Ubuntu.

## Onboard eMMC

The onboard eMMC uses MMC1 signals. It is also connected to the expansion header. The eMMC device on the Beaglebone Black will need to be disabled by activating the reset line to this device. To do this, you have to write to eMMC first and instruct it to enable the reset. At this point in time, it is not totally clear whether the reset line will do

the trick. The best option is not to use the MMC1\_CLK and MMC1\_CMD signals at all and tie them 'LOW'.

**Note:** If the board is powered off without using the onboard power button, then there is a chance of data corruption in the eMMC. The data in the eMMC can be restored by re-flashing the board. It is always recommended to use the latest version of the software.

## Steps to flash the board



Figure 1: Beaglebone Black

To flash the board, you need an SD card with a capacity of 4GB or more. Using an SD card smaller than 4GB will not boot the board. Before starting, do remember to take a backup of all the contents of the SD card that you are going to use, as flashing will erase them completely. Now, ensure to download the latest images for the Beaglebone Black from the official Beagleboard website (Figure 2).

The image can also be directly downloaded from the system by using the following command:

```
$ wget -c http://debian.beagleboard.org/images/BBB-eMMC-
```



Figure 2: Latest image source